

Response to Reviewers

Energy-Efficient Deep Learning Without Backpropagation:
A Rigorous Hardware-Validated Benchmarking Study of Forward-Only Algorithms
PPAM 2026, Submission 25 (CP025)

Summary

We thank all five reviewers. Two of the comments did more than improve the paper: they produced the experiment that the revised version is built on, and they changed our headline result. We start with that one.

R3c and R5d asked the decisive question, and the answer is in MF’s favour. Both reviewers observed that MF places a cross-entropy loss at every hidden layer, so it behaves like a deep-supervision network, and that without a matched deep-supervision baseline we could not tell whether MF’s accuracy comes from forward-only training or merely from localised objectives. That control now exists. BP-DS is BP carrying MF’s auxiliary classifiers with nothing detached, and on the two CIFAR 3×2000 MLPs it is *worse* than plain BP, by 1.65 pp on CIFAR-10 and 1.06 pp on CIFAR-100. MF beats that control by 5.56 pp [5.37, 5.75] and 6.10 pp [5.88, 6.32] over 20 paired seeds. The localised objective does not explain the gain; adding it to BP lowers BP’s accuracy. The evidence therefore isolates gradient detachment/locality as the dominant tested factor; we did not test every possible mechanism. On the two MNIST-scale 2×1000 MLPs the same contrast is a tie, and we report that half too. The full four-rung ladder is Section 5.3, Table 3 and Figure 2 of the revised paper.

R2d asked whether BP simply has better kernels, and the telemetry answers it. We tested the objection rather than arguing it. On the four ladder configurations no method exceeds 20.3 % mean GPU utilisation and BP itself averages 11.5 %; across the whole study, including the wider MLPs used for FF, no run exceeds 38.0 %. Every arm is launch-bound, not arithmetic-bound, so BP does not win by saturating the A100 with more efficient arithmetic. We also built the obvious missing optimisation, an activation cache that removes MF’s $O(L^2)$ recomputation, verified it bit-exact against the recomputing path (maximum absolute difference in loss and weights exactly 0 over 1200 and 3600 optimiser steps), and measured that it saves no time while costing 374 % and 404 % more per-process allocator memory. These measurements do not entirely exclude an advantage from implementation maturity, but the difference they do measure between the methods lies in convergence rate rather than in the cost of a pass.

The cost result has two readings, and this version states both. Trained to its own convergence, MF costs $3.2\times$ to $4.0\times$ BP’s energy. The reason it costs more end to end is that it needs $3.6\times$ to $4.4\times$ the wall-clock time, not that each pass is expensive. Given the number of data passes BP actually spent, MF is the *cheaper* algorithm: 19.9 % less GPU energy on CIFAR-10 and 12.5 % less on CIFAR-100, 27.5 % and 24.6 % less per-process allocator memory, at wall-clock time within 3 % of BP’s. The revised paper leads with that matched data-pass protocol result (Section 5.1, Table 2), keeps the own-convergence result in full (Section 5.4, Table 4), and names the pass multiplier as the open problem (Section 6).

What is new in this version, concretely.

- **A matched data-pass protocol**, in which MF receives $\lceil \bar{E}_{BP}/S \rceil$ epochs per stage, so rounding can only give MF more data passes than BP had. The protocol equalises data passes, not

floating-point operations or optimiser updates per layer, and is named accordingly. Section 4, Table 1.

- **A pre-registered four-rung ablation ladder** from BP to MF, including the BP-DS control of R3c and R5d, over 20 to 122 seeds per rung. Section 5.3.
- **A new figure showing that the advantage is larger in the larger configurations tested**, from no memory advantage at 1.8 M weights to 27.5 % at 14.2 M. Figure 1.
- **Seed counts of 20 to 122 per MF contrast** in place of three, with every parity claim resolved by the two one-sided tests (TOST) procedure against a pre-registered ± 0.25 pp margin, now cited and explained in Section 4. R4b, R5c.
- **Every figure regenerated** from the measurement archive by a script in the repository. The illegible dashboard exports of R1b are gone. R1b, R3b.

Every measured number in the revised paper is produced from a single tidy table of 19944 measurements by `analysis/analyse.py`, which is in the repository alongside the figure script; only the realised epoch counts are transcribed, from the run configurations. The manuscript is **15 pages**, the PPAM limit.

Revised structure. §1 Introduction | §2 Background and Related Work | §3 Mono-Forward and the Cost Hypothesis | §4 Measuring Cost Fairly: Two Protocols | §5 Results (5.1 the matched data-pass protocol, 5.2 scale, 5.3 the ladder, 5.4 own convergence, 5.5 quantifying two objections, 5.6 FF and CaFo) | §6 Discussion and Limitations | §7 Conclusion. Table 1 = budgets; Table 2 = matched data-pass protocol; Table 3 = the ladder; Table 4 = own convergence. Figure 1 = saving against model scale; Figure 2 = the ladder.

Reviewer 1

R1a: “Reads like a very condensed technical report . . . detrimental to readability for the average PPAM reader.”

Reads like a very condensed technical report, which is detrimental to readability for the average PPAM reader.

Accepted, and the paper has been rewritten rather than edited. The accepted version was organised as one subsection per algorithm, which forced the reader to assemble the argument themselves. The revision is organised around one claim and its evidence: Section 3 states a mechanism and why it should be cheaper, Section 4 states the two protocols and why two are needed, Section 5 tests the claim, and Section 6 says where it fails. Three of the six results subsections (5.1, 5.2 and 5.4) carry a verdict in their titles, and the three emphasised sentences of Section 5.1 (*the memory figures are exact, the energy saving is a power saving, on the most demanding task the saving costs no accuracy*) give a reader the result without the table.

Two of the narrowing devices are deliberate. We dropped the algorithm-by-algorithm survey structure and compressed FF and CaFo into a single subsection (5.6), because the study’s defensible contribution is about MF and the other two are context for it. We also state the scope of every headline number in the sentence that carries it, rather than deferring all qualification to a limitations section.

Where: Whole manuscript; Sections 3 to 6 are new text.

R1b: “The plots in Fig. 3 and Fig. 4 are completely illegible.”

The plots in Fig. 3 and Fig. 4 are completely illegible.

Accepted. Those figures were raw dashboard exports at a size no print process could rescue. They have been deleted, not enlarged.

The revision’s figures are generated by `analysis/figures.py` from the same tidy table as the tables, at the text width of the `llncs` block, with every label in Figures 1 and 2 at 8 pt or larger and no element carrying information by colour alone. Series are separated by fill, hatch and line style together, so the figures survive a monochrome print, which we checked by rendering them in greyscale. Figure 1 is a four-group bar chart and Figure 2 is two line panels over four ladder rungs; neither has more than four series. A third figure showing both protocols side by side is generated by the same script and is in the repository, left out of the manuscript only for space, since Table 4 carries the same numbers.

Where: Figures 1 and 2 and their captions; the supplementary third figure; `analysis/figures.py`.

Reviewer 2

R2a: “Lack of details and analysis in the main article (specifically Section 6) made the paper seem preliminary.”

Lack of details and analysis in the main article (specifically Section 6) made the paper seem preliminary.

Accepted. The discussion is now built from measurements rather than from interpretation, and it names what it cannot support.

It is built in two steps, followed by the implications; the robustness of the cost result (the matched data-pass protocol gives MF at least as many data passes as BP used, and board power is a rate that no stopping rule can influence) is stated where it applies: the budget rule in Sections 1 and 4, the power argument with the result itself in Section 5.1. *Why gradient detachment might improve accuracy* advances the regularisation reading, ties it to the ladder result, and labels it a hypothesis that four configurations cannot establish. *The open problem: convergence rate, not per-unit cost* identifies convergence rate as the binding constraint, shows that the two obvious remedies fail (caching is bit-exact and saves no time; the runs are launch-bound so faster kernels would be an unlikely remedy, although implementation maturity is not entirely excluded), and bounds the result on three further axes: scope, accuracy under the matched data-pass protocol, and hardware. What was pre-registered and what was not is stated in Section 4.

We also state plainly where the measurements and the narrative pulled against each other. Under the matched data-pass protocol MF’s energy is lower on all four configurations by point estimate, a difference resolved on three, but it is as accurate as BP on only one of the four; we say so in Section 5.1 and repeat it in the limitations, because it bounds the claim.

Where: Section 6, all paragraphs.

R2b: “The details of the different ML architectures are important ... these details are not discussed much in the paper.”

The details of the different ML architectures are important, as they can affect the results; these details are not discussed much in the paper.

Accepted, and the revision goes further than describing them: it makes architecture an axis of the result.

Section 3 gives MF’s construction explicitly, including the fact that an MLP with L hidden layers trains $L + 1$ stages, one per projection matrix including $\mathbf{M}_0$ on the raw input, which is what sets the epoch budget in Table 1. Section 2 now introduces the notation and briefly describes FF, CaFo and MF before they are used. Section 4 states initialisation (Kaiming uniform for all weights and projections, now with references for it and for both optimisers), the optimiser per algorithm (AdamW for BP and BP-DS, Adam for MF-Joint and MF stages), the activation (ReLU) and batch size (128), the tuning budget and search width per algorithm, and the augmentation. It records that every algorithm is built on the same `nn.Linear` primitives in one codebase, differing only in the

loss construction and in where `detach()` is called. The remaining per-algorithm implementation constants, the CaFo block (3×3 convolution, ReLU, 2×2 max-pooling, batch normalisation, 32/128/512 channels, three block predictors summed) and the FF goodness threshold $\theta = 2.0$ with peer normalisation, are in the released configurations, since they bear on FF and CaFo rather than on the claim.

Section 5.2 then uses the architectures. The four configurations span 1.79 M to 14.34 M trainable weights, and every cost axis improves across that range: energy saving from 2.2 % and 4.3 % to 12.5 % and 19.9 %, memory from a 0.3 % deficit to a 24.6 % and 27.5 % saving, so the advantage is larger in the larger configurations tested. We are explicit that width is confounded with depth and with task across these four, so the figure cannot separate width from scale, and Section 6 states that the headline percentages hold on the 3×2000 MLPs only.

Where: Section 3; Section 4, “What else is held fixed”; Section 5.2 and Figure 1.

R2c: “The paper claimed that the BP baselines were identical but algorithm-specific components were removed. The removal of these components may be the reason why certain architectures were better one way or another.”

The paper claimed that the BP baselines were identical but algorithm-specific components were removed. The removal of these components may be the reason why certain architectures were better one way or another.

This is correct, it was a real hole, and the ladder was built to close it. We did not argue that the removal was harmless; we put the components back one at a time and measured what each is worth.

Rung	Readout	Auxiliary losses	Gradients	Optimiser
BP	output layer	no	global	AdamW
BP-DS	output layer	yes	global	AdamW
MF-Joint	$\mathbf{M}_L$	yes	global	Adam
MF	$\mathbf{M}_L$	yes	local	Adam

Each adjacent pair differs in one structural property. We should be precise about what that does and does not license: each rung is tuned independently under the same 50-trial budget, so every step also moves the learning rate and weight decay, and the BP-DS→MF-Joint step additionally crosses AdamW to Adam. That step is therefore not attributable to the readout alone, and we no longer claim it is. The step our mechanism claim rests on, MF-Joint→MF, holds the optimiser family fixed at Adam. Accuracy against BP, in percentage points, as a difference of arm means over 20 to 122 seeds:

	MNIST 2×1000	Fashion 2×1000	CIFAR-10 3×2000	CIFAR-100 3×2000
BP-DS	+0.07	+0.04	−1.65	−1.06
MF-Joint	+0.14	−0.09	−0.19	+1.86
MF	+0.19	+0.01	+3.90	+5.04

Adding the auxiliary classifiers to BP does not explain MF’s advantage; on the two CIFAR tasks it costs BP accuracy. The readout step recovers most or all of the loss, +1.46 and +2.92 pp, but it is the step that also crosses optimisers, so we do not attribute it to the readout. The largest single step, and the only one we attribute to a mechanism, is the detach, worth +4.10 pp on CIFAR-10 and +3.18 pp on CIFAR-100 with the optimiser held fixed. The same step also carries essentially the whole cost increase, 228 to 272 pp of BP’s energy, while the other two steps together move energy by at most 31 pp.

Where: Section 5.3, Table 3, Figure 2; per-contrast intervals and Holm-corrected p -values in `analysis/ladder_analysis.json`, released with the paper.

R2d: “As backprop algorithms are more widely used, the functionalities in existing software libraries could have been better optimised ... than forward-only algorithms.”

As backprop algorithms are more widely used, the functionalities in existing software libraries could have been better optimised for backprop than for forward-only algorithms.

This is the objection we most wanted to be able to answer with data, and the utilisation trace answers it. If kernel maturity decided the comparison, we would expect BP to be near saturation and the forward-only methods starved of it. Neither holds. On the four ladder configurations, mean GPU utilisation is 11.5 % for BP (maximum 17.3 %) and 6.0 % for MF (maximum 8.0 %), and no method exceeds 20.3 %. Across the entire study, the wider FF MLPs included, no run exceeds 38.0 % mean utilisation, still far from saturation. Every arm is bound by kernel launch latency, not by arithmetic throughput. This shows that BP does not win by saturating the A100 with more efficient arithmetic operations. It does not entirely exclude an advantage from implementation maturity, and the revised paper says so in Sections 5.5 and 6.

We also tested the specific optimisation a mature implementation would be expected to have. MF’s layer-sequential schedule recomputes the frozen prefix at every stage, an $O(L^2)$ cost that an activation cache removes, so we implemented the cache. It is bit-exact against the recomputing path: maximum absolute difference in both loss and weights exactly 0 over 1200 and 3600 optimiser steps. It bought no time (+3.8 % and +2.7 % wall-clock on device) and cost 374 % and 404 % more per-process allocator memory, which is 242 MiB and 257 MiB against 51 MiB. Caching in host memory was worse on every axis. The optimisation is available, and on this hardware it does not help.

Finally, the direction of the result does not need the objection resolved. Under the matched data-pass protocol MF is the *cheaper* method, so any library-maturity advantage BP holds is working against the conclusion we draw, not for it.

Where: Section 5.5, both paragraphs.

Reviewer 3

R3a: “NVML ... is never formally defined or explained.”

NVML is used throughout but is never formally defined or explained.

Accepted. NVML is now expanded and defined on first use, in the introduction, as the NVIDIA Management Library, the driver’s own telemetry interface, and the expansion is repeated in Sections 2 and 4. Section 4 has a dedicated *Instruments* paragraph naming the exact API call behind every quantity: `nvmlDeviceGetPowerUsage` for power, from which energy is the trapezoidal integral of the 5 Hz trace; `nvmlDeviceGetUtilizationRates` for utilisation; `nvmlDeviceGetClockInfo` for SM clock; `nvmlDeviceGetMemoryInfo` for device memory; and the host monotonic clock for wall-clock time.

The same paragraph separates the two memory figures that the accepted version conflated. The NVML reading is device-wide and includes a ≈ 870 MiB CUDA context that no algorithm controls. Every memory number in the revised paper is instead `torch.cuda.max_memory_allocated`, the per-process allocator high-water mark, and we say so at each use. The driver version is also stated, in “Setup and statistics”, because it governs NVML power-reading semantics.

Where: Sections 1 and 2; Section 4, “Instruments” and “Setup and statistics”.

R3b: “The results for CaFo in Section 5.2 are presented purely as text ... should be summarised in an equivalent table.”

The results for CaFo in Section 5.2 are presented purely as text, and should be summarised in an equivalent table.

Accepted in substance, but we did not do exactly what was asked. To remain within the 15-page limit, we replaced the requested table with a compact numerical summary in Section 5.6 and provide the complete table in the repository. The revision is built around a single claim about MF, and FF and CaFo are now context for it rather than co-equal subjects. Giving each of them a results table would have cost roughly two pages of a 15-page limit for material that does not bear on the claim.

Instead, Section 5.6 states every CaFo and FF result numerically rather than qualitatively: for FF, the accuracy gap to BP across its four arms and its time, energy and memory as multiples of BP’s; for CaFo, the accuracy gap on the MNIST-scale and CIFAR tasks and its energy as a multiple of BP’s; and the seed counts. CaFo costs $3.0 \times$ to $7.1 \times$ BP’s energy, lands within 0.4 pp of BP on the two MNIST-scale CNNs and loses 18.8 and 16.9 pp on CIFAR-10 and CIFAR-100; the direct-feedback-alignment variant narrows that to 7.1 and 8.5 pp at a further $1.5 \times$ to $2.6 \times$ the energy. The complete per-configuration table for both algorithms is released as `tables/ff_cafo_full.csv`, generated by `analysis/ff_cafo_table.py` from `analysis/numbers.json`, which the released analysis computes from the tidy table cited in the paper.

If the editors prefer the full table in the paper we will add it and cut Section 5.6’s prose to a cross-reference, which is close to page-neutral.

Where: Section 5.6; in the repository, `tables/ff_cafo_full.csv`, `analysis/numbers.json` and `artifacts/tidy/runs.csv`.

R3c: “Because Mono-Forward places a cross-entropy loss at every hidden layer, it inherently behaves like a deep supervision network ... add a standard backpropagation baseline that uses matching layer-wise auxiliary losses. This is the only way to prove whether MF’s gains stem from its unique forward-only mechanism or simply from localised objective constraints.”

Because Mono-Forward places a cross-entropy loss at every hidden layer, it inherently behaves like a deep supervision network. Add a standard backpropagation baseline that uses matching layer-wise auxiliary losses. This is the only way to prove whether MF’s gains stem from its unique forward-only mechanism or simply from localised objective constraints.

This is the most valuable comment we received and we have acted on it in full. BP-DS is exactly the baseline described: BP with MF’s auxiliary classifiers attached at every hidden layer and nothing detached, trained under the same stopping rule, the same tuning budget and the same seeds.

The answer is decisive on the harder tasks and honest on the easier ones. MF beats BP-DS by 5.56 pp [5.37, 5.75] on CIFAR-10 and 6.10 pp [5.88, 6.32] on CIFAR-100, both over 20 paired seeds with Holm-corrected $p < 10^{-20}$. BP-DS is meanwhile *worse* than plain BP on those two tasks, by 1.65 and 1.06 pp. Localised objectives therefore do not explain the gain; on the tasks where MF gains most, adding them to BP costs accuracy. On the two 2×1000 MLPs the contrast is a tie, MF leading BP-DS by 0.11 pp [−0.01, 0.23] on MNIST and trailing by 0.06 pp [−0.17, 0.06] on Fashion-MNIST, both EQUIVALENT within the pre-registered margin. We report both halves in the paper: on the small tasks the experiment does not distinguish an MF-specific benefit from auxiliary supervision, and on the harder ones auxiliary supervision does not explain it.

Two further points on the strength of this evidence. The ladder separates the readout from the detach as well, using a third rung, MF-Joint, which has MF’s $\mathbf{M}_L$ readout and auxiliary losses but a global gradient. The step from MF-Joint to MF detaches the gradient with the optimiser family held fixed and is worth +4.10 pp on CIFAR-10 and +3.18 pp on CIFAR-100. And the whole analysis is pre-registered: the script fixing the paired test, the Holm correction within metric family, the bootstrap intervals and the ± 0.25 pp margin was committed before any ladder result was inspected.

Where: Section 5.3, Table 3, Figure 2; `scripts/analyze_ablation_ladder.py` for the pre-registration.

R3d: the memory finding highlighted in the reviewer’s summary, and how to read the revision against it

Crucially, the study debunks a common assumption in BP-free literature by showing that the actual memory overhead introduced by algorithm-specific components (such as local projection matrices) frequently counteracts or entirely offsets the theoretical memory savings gained from omitting the backward pass.

The reviewer singled this out, the revision now headlines a 24.6 % to 27.5 % memory *saving* for MF, and we want to be explicit that this is a refinement of the finding rather than a retraction of it.

The finding stands and the revision still carries it. FF holds $0.91 \times$ to $1.06 \times$ BP’s allocator peak, so its dual positive and negative data streams cancel most or all of the saving (Section 5.6). MF at 1.8 M weights shows a 0.3 % *deficit*, which is exactly the 0.14 MiB of extra projection parameters outweighing what the absent backward pass saves (Section 5.2). What the revision adds is the other end of the same axis: at 14.2 M weights the activation saving outgrows the projection overhead and the sign flips. The reviewer’s point was that the overhead is real and must be measured rather than assumed, and the new Figure 1 is that point plotted, from a deficit at the small size to a 27.5 % saving at the large one.

One correction belongs here. The accepted version’s 5 % memory figure came from the device-wide NVML counter, in which a ≈ 870 MiB CUDA context dilutes every difference. Every memory number in the revision is `torch.cuda.max_memory_allocated`, the per-process allocator high-water mark. The instrument changed; the conclusion the reviewer identified did not.

Where: Section 5.2 and Figure 1; Section 5.6; Section 4, “Instruments”.

Reviewer 4

R4a: “The key term ‘hardware-validated comparison’ is unclear and should be explained in more detail.”

The key term “hardware-validated comparison” is unclear and should be explained in more detail.

Accepted. The revision does not rely on the phrase to carry the meaning. The introduction states what is measured and how, the *Instruments* paragraph of Section 4 names the API call behind each quantity, and Section 2 explains why this matters rather than asserting that it does: analytical proxies such as FLOP or parameter counts cannot see clock behaviour, utilisation or kernel launch overhead, which is precisely where the differences in this study turn out to live.

Section 5.5 makes the point concrete. A FLOP-based comparison of MF and BP would predict a speed-up from the cache and see no difference in power. The device says the opposite: the cache is bit-exact and saves no time, and the entire energy result is a power difference at near-equal wall-clock time. That gap between the proxy and the instrument is what “hardware-validated” is doing in the title.

Where: Section 1; Section 2, “Hardware and energy”; Section 4, “Instruments”; Section 5.5.

R4b: “The statistical analysis is based on only three runs, which is insufficient and can be considered only as preliminary evidence.”

The statistical analysis is based on only three runs, which is insufficient and can be considered only as preliminary evidence.

Accepted without reservation. Three seeds was insufficient and we have not tried to defend it.

Every MF contrast in the revised paper is over at least 20 seeds (the FF and CaFo context runs use five to seven), and the two most heavily used configurations carry 60 and 122. The matched data-pass protocol comparison of Table 2 uses 20 MF seeds against 60, 122, 20 and 20 BP seeds

in column order, paired by seed where the arms share one. The ladder of Table 3 uses 20 to 122 seeds per rung. The study as a whole is almost 1000 runs.

The analysis itself changed as much as the seed count. The primary test is paired by seed, because a seed fixes both the train and validation split and the initialisation. A Welch unequal-variance test is computed alongside it in the released analysis, never instead of it. Holm-Bonferroni correction is applied within each metric family and never pooled across families. Confidence intervals come from 10 000 bootstrap resamples with a fixed seed. Most importantly, no parity claim rests on a non-significant difference test: every one uses the two one-sided tests (TOST) procedure, now cited and explained in Section 4, against a ± 0.25 pp margin fixed in advance, and the verdict vocabulary is restricted to DIFFERENT, EQUIVALENT and INCONCLUSIVE.

One scope statement belongs here. The ladder analysis is pre-registered, and the paper says so. The matched data-pass protocol is *not* pre-registered; it is analysed with the same machinery and the same margins, but the paper labels it a diagnostic rather than a confirmatory test.

Where: Section 4, “Setup and statistics”; seed counts in the captions of Tables 2 to 4.

Reviewer 5

R5a: “The contribution is mainly evaluative rather than algorithmic.”

The contribution is mainly evaluative rather than algorithmic.

The observation is accurate and we have not tried to dress the paper up as something else. What we have done is make the evaluative contribution carry a mechanistic result rather than a league table.

The ablation ladder is the difference. It does not report that MF differs from BP; it decomposes the difference into three single structural steps and identifies the same one, the detach, as the largest contributor to both the accuracy and the cost among the factors tested. That attribution is falsifiable, it was pre-registered, and it contradicts the reading a reader would most naturally have taken, namely that MF’s auxiliary losses do the work. It also produces a negative result of its own: deep supervision on top of BP is worse than BP on both CIFAR tasks.

The second non-obvious contribution is methodological. The same algorithms support opposite cost conclusions depending on the stopping rule, and the revision shows both and quantifies the distance between them. We think the general statement is worth making: an efficiency claim about a training algorithm is uninterpretable without its stopping rule, because the rule is not a detail of the measurement but a term in it. That is stated in Section 6 as a claim about the field, not about this algorithm.

Where: Section 5.3; Section 6, “Implications”.

R5b: “Small datasets, shallow architectures, and the use of flattened CIFAR inputs for MLP experiments.”

Small datasets, shallow architectures, and the use of flattened CIFAR inputs for MLP experiments.

Accepted as a limitation, stated as one, and partly turned into a result.

On flattening: CIFAR inputs are flattened because that is the protocol of the MF source publication, and departing from it would make our numbers incomparable with the ones we are testing. The important point is that flattening is applied identically to MF and to its BP baseline, along with the same augmentation, so it cannot favour either. Section 4 states both: flattening follows the MF protocol for the MLP experiments, and the augmentation is applied identically to every algorithm and its baseline.

On scale: the revision states the bound rather than hoping the reader supplies it. The headline percentages hold on the 3×2000 MLPs, no convolutional MF was run, and both statements appear

in Section 6. What we can offer instead of a larger model is the direction of travel, measured: across a nearly eightfold range of trainable weights every cost axis improves, and the memory advantage exists only at the larger size, so the advantage is larger in the larger configurations tested, with size confounded with depth and task. If the effect scaled the other way, that would be visible in Figure 1, and it is not.

We would rather report a trend over the range we could afford to run almost 1000 times than a single point at a larger scale with three seeds.

Where: Section 4; Section 5.2 and Figure 1; Section 6, limitations.

R5c: “Results are based on only three runs, and some of the reported accuracy differences are modest.”

Results are based on only three runs, and some of the reported accuracy differences are modest.

The seed count is addressed under R4b: 20 to 122 per MF contrast, almost 1000 runs in total.

The second half deserves its own answer, because the reviewer is right that several differences are small, and the revision treats small differences as such. Where a difference is modest we no longer call it a win. Every accuracy parity claim is a two one-sided test against a ± 0.25 pp margin fixed before the data were seen, and a contrast whose interval does not fit inside the band is marked INCONCLUSIVE rather than being read as parity. That is why the paper reports MF as equivalent to BP under the matched data-pass protocol on CIFAR-100 (+0.02 pp, EQUIVALENT) but as 1.11 pp *behind* on CIFAR-10 (DIFFERENT), and does not average the two into a claim of parity.

The differences the paper does lean on are not modest. MF over BP-DS is 5.56 and 6.10 pp; MF over BP at its own convergence is 3.90 and 5.04 pp; the memory saving is 82.2 and 74.4 MiB with zero seed-to-seed variance, because the allocation schedule of a fixed architecture at a fixed batch size is deterministic.

Where: Section 4, “Setup and statistics”; Section 5.1; Section 5.3.

R5d: “A stronger comparison would include additional baselines, such as deep-supervision or local-loss backpropagation variants, to better isolate whether the gains come from the forward-only training mechanism or from local auxiliary objectives.”

A stronger comparison would include additional baselines, such as deep-supervision or local-loss backpropagation variants, to better isolate whether the gains come from the forward-only training mechanism or from local auxiliary objectives.

Done, and the answer is the one reported under R3c and in the Summary above. Two additional baselines were built, tuned and run at 20 to 122 seeds each:

BP-DS

deep supervision: BP with MF’s layer-wise auxiliary classifiers, nothing detached. This is the local-auxiliary-objective control the reviewer asks for.

MF-Joint

MF’s learned $\mathbf{M}_L$ readout and auxiliary losses with a global gradient. This separates the extra parameters from the detach, so the readout cannot be credited with the locality effect.

The separation is clear on the CIFAR tasks. Auxiliary objectives alone lose 1.65 and 1.06 pp against plain BP on those two tasks. Adding the readout recovers most or all of that, reaching -0.19 and $+1.86$ pp. Only the final rung, which adds the detach, reaches $+3.90$ and $+5.04$ pp. Among the factors tested, the gain is therefore attributable mainly to gradient detachment rather than to the local objectives, which on these two tasks lower accuracy when used alone.

Where: Section 5.3, Table 3, Figure 2.

R5e: “The energy measurements are useful, although specific to one GPU platform.”

The energy measurements are useful, although specific to one GPU platform.

Agreed, and the revision states the dependence in the specific form we think it takes, rather than as a generic caveat.

All runs are on NVIDIA A100-SXM4-40GB with driver 595.71.05, and the driver version is stated because it governs NVML power-reading semantics. The limitation is not merely that another device might give another number; it is that we can say which way. Board power at these utilisations is dominated by the static and clock-domain components of a large A100 die, and the entire energy result of Table 2 is a power difference at near-equal wall-clock time. A smaller or better-saturated device would very likely narrow that power gap and so narrow the energy saving. Section 6 says exactly this.

Two components of the result are more portable than the energy figure. The memory saving is an allocator high-water mark and follows from the algorithm’s activation lifetime rather than from the device, and the pass multiplier that dominates the own-convergence comparison is a property of the optimisation, not of the hardware. We would expect both to survive a change of accelerator.

Where: Section 4, “Setup and statistics”; Section 6, limitations.

R5f: “Clarify the limitations, and strengthen the discussion of baselines and statistical robustness.”

Clarify the limitations, and strengthen the discussion of baselines and statistical robustness.

All three are addressed above and consolidated in Section 6, which now names the binding limitation rather than listing caveats. That limitation is convergence rate: MF needs $3.6 \times$ to $4.4 \times$ BP’s wall-clock time to reach its own best accuracy, which converts a per-pass advantage of 12.5% to 19.9% into an end-to-end penalty of $3.2 \times$ to $4.0 \times$. The paper shows that the two obvious remedies do not close it and states it as the open problem, which is where we think work on forward-only efficiency should go.

Three further limitations are bounded explicitly in Section 6: scope (the headline percentages hold on the 3×2000 MLPs, and there is no memory advantage at 1.8 M weights); accuracy under the matched data-pass protocol (equivalent on one configuration of four, and 1.11 pp behind on CIFAR-10, so the matched data-pass protocol result is a cost result and not an accuracy result); and hardware (R5e). Pre-registration (the ladder is, the matched data-pass protocol is not) is stated in Section 4. Baselines are R2c, R3c and R5d. Statistical robustness is R4b and R5c.

Where: Section 6, “The open problem: convergence rate, not per-unit cost”.

We are grateful for the review. R3c and R5d in particular produced the experiment that this version is built on, and it is a better paper than the one that was accepted.